Stability and comparability of meritocratic beliefs in school-age students: A measurement invariance approach across time and cohorts

Andreas Laffert[1], Juan Carlos Castillo[2], René Canales[3], Tomás Urzúa[4], Kevin Carrasco[5]

[1]Institute of Sociology, Pontifical Catholic University of Chile
[2]Department of Sociology, University of Chile
[3]Institute of Sociology, Pontifical Catholic University of Chile
[4]Department of Sociology, University of Chile
[5]Department of Sociology, University of Chile

Abstract

Meritocratic beliefs shape how adolescents understand inequality and the legitimacy of social arrangements. Schools are key sites of meritocratic socialization, making adolescence a crucial period for studying how these beliefs are formed. However, research on adolescents' meritocratic beliefs has often relied on unidimensional measures that conflate perceptions of how allocation works with preferences about how it should work and treat meritocratic and non-meritocratic principles as opposites. This study tests a multidimensional model that distinguishes between perceived and preferred meritocracy and non-meritocracy. Using two-wave panel survey data from Chilean students in 8th and 10th grade (Wave 1: N = 846; Wave 2: N = 662), we assess whether these dimensions can be distinguished, compared across school stages, and measured equivalently over time. Results support a four-factor structure and show that endorsement of merit can coexist with recognition of non-meritocratic advantage. The scale achieves strict longitudinal invariance within students but not across school levels.
Keywords: Meritocracy · Schools · Socialization · Measurement invariance · Chile

1 Introduction

Despite rising economic inequality and limited social mobility in contemporary societies (Chancel et al., 2025; López-Roldán & Fachelli, 2021), the belief that inequalities reflect disparities in effort and talent instead of opportunities remains remarkably widespread among citizens (Mijs, 2021). This idea refers to meritocracy, the notion that social rewards are, and should be, allocated according to individual effort and ability rather than social origins or personal connections (Bell, 1972; Young, 1958). Although the term meritocracy was originally coined as a critique of a future society where meritocratic principles would be taken to an extreme (Young, 1958), it has since been widely adopted as a normative ideal that justifies existing social arrangements and motivates individual striving (Mijs, 2021; Sandel, 2020). On the one hand, meritocracy can be seen as a powerful narrative that promotes social cohesion and motivates individuals to work hard and pursue their goals. On the other hand, it has been largely criticized for obscuring structural inequalities and legitimizing the status quo (Hoyt et al., 2023; Tejero-Peregrina et al., 2025). Therefore, the persistence of meritocratic beliefs could have important implications for how people understand and respond to inequality, as well as for the legitimacy of social institutions (Madeira et al., 2019; Sandel, 2020).

The idea of meritocracy is particularly salient in schools, where it operates both as a cultural ideal and as an organizing principle of institutional life. Schools promote meritocracy by presenting education as a legitimate route to upward mobility and individual effort and achievement as the proper basis of social rewards (Chauvin et al., 2026; Darnon, Wiederkehr, et al., 2018; Mijs, 2016b). Students encounter this ideal through discourse as well as grading, selection, and tracking, which make merit visible and consequential (Darnon et al., 2023; Resh & Sabbagh, 2017; Traini, 2022). At the same time, they are exposed to non-meritocratic factors, such as family background, social connections, and structural barriers, which may coexist with or undermine the meritocratic ideal (Bourdieu & Passeron, 1990; Darnon et al., 2023; Goudeau & Croizet, 2017; Tang et al., 2025). Stronger endorsement of school meritocracy has been linked to greater justification of inequality (Batruch et al., 2022; Castillo et al., 2024; Darnon, Smeding, et al., 2018; Darnon, Wiederkehr, et al., 2018; Wiederkehr et al., 2015), even when students recognize structural constraints, reflecting a form of dual consciousness (C. Liu & Wang, 2025; Tang et al., 2025). These tensions make schools a crucial setting for examining how adolescents understand meritocracy and the fairness of unequal outcomes.

The present paper aims at contributing to studying how school-aged students perceive and endorse meritocracy, with two main aims. The first one is the conceptualization and measurement of meritocratic beliefs in school settings. Most of the research in this area has focused on the consequences of meritocratic beliefs, while less attention has been paid to their conceptualization and measurement. This is a crucial gap in the extant literature, as how meritocratic beliefs are conceptualized and measured can have important implications for the validity of research findings and their interpretation. Addressing this point, we approach the study of meritocratic beliefs at school from a multidimensional approach (Castillo et al., 2023), which distinguishes between perceptions (descriptive beliefs about how meritocracy works) and preferences (normative beliefs about how it should work), as well as between meritocratic principles (e.g., effort and ability) and non-meritocratic principles (e.g., family background and social connections). By adopting this perspective, we can better capture the complexity of adolescents' beliefs about meritocracy and the role schools play in fostering them.

A second aim of this study is to examine the stability and comparability of meritocratic beliefs across school stages and over time. Most research on these beliefs remains cross-sectional, limiting our understanding of how they develop through school-based socialization during adolescence. This is especially important because adolescence is when reasoning about justice and inequality becomes more sophisticated and school-based sorting mechanisms become salient (Henry & Saul, 2006; Resh & Sabbagh, 2014, 2017), potentially reshaping how students interpret meritocratic narratives in light of lived constraints (Tang et al., 2025). For this, we assess the equivalence of the multidimensional model of meritocracy across two cohorts of Chilean students (8th and 10th graders) and longitudinally within the same students across two waves (2023–2024). Establishing this measurement-invariance foundation is essential both for distinguishing developmental change from measurement non-equivalence (Putnick & Bornstein, 2016; Van De Schoot et al., 2015) and for examining the consequences of meritocratic beliefs for attitudes towards inequality.

According to these two aims, the research questions guiding this study are as follows:

(1) Do school-aged students distinguish between meritocratic and non-meritocratic principles, as well as between perceptions ("what is") and preferences ("what should be")?

(2) Can these dimensions be measured in ways that are comparable across age cohorts and stable over time?

By addressing these questions, this study contributes to a better understanding of how meritocratic beliefs are structured and how they develop during adolescence, shedding light on the role of schools in shaping these beliefs and their implications for attitudes and behaviors toward inequality and achievement in both society and educational settings.

2 Theoretical and empirical background

2.1 What is meritocracy, and how is it measured?

Meritocracy refers to a distributive system in which individual merit—typically effort and ability—is treated as the main criterion for allocating rewards, rather than social origins or inherited privilege (Bell, 1972; Young, 1958). Although Young (1958) introduced the term as a dystopian critique, it has since been reappropriated as a positive ideal of fairness, especially in liberal and market-oriented societies (Mijs, 2021; Van De Werfhorst, 2024). Sociologically, meritocracy operates both as a cognitive judgment about how inequality works and as a moral lens through which inequality is evaluated (Castillo et al., 2019; Heuer et al., 2020). As meritocracy frames outcomes as earned, inequalities could be seen as legitimate: winners see their position as deserved and losers to internalize failure rather than question structural conditions (García-Sierra, 2023; Mijs, 2016a; Sandel, 2020).

Research on adults has examined both the determinants and consequences of meritocratic perceptions. Higher-status and upwardly mobile individuals are more likely to endorse merit-based explanations of social outcomes (Duru-Bellat & Tenret, 2012; García-Sánchez et al., 2018; Mijs et al., 2022; Traini et al., 2025), whereas perceiving society as meritocratic is associated with lower support for redistribution, greater acceptance of inequality, and stronger endorsement of market-based allocation (Castillo et al., 2025, 2019; Hoyt et al., 2023; Pañeda-Fernández et al., 2026; Tejero-Peregrina et al., 2025). Experimental evidence is consistent with a causal pathway in which upward-mobility or self-made narratives strengthen meritocratic perceptions and reduce redistributive support, while downward-mobility cues produce the opposite pattern (Deng & Wang, 2025; Matamoros-Lima et al., 2025). Overall, this literature suggests that meritocratic perceptions are systematically linked to both social position and orientations toward inequality.

The research agenda on meritocratic beliefs faces important conceptual and measurement challenges. Some studies define meritocracy narrowly, often reducing it to effort or simplified readings of Young's (1958) formulation (Day & Fiske, 2017; Mijs, 2021; Wiederkehr et al., 2015), while others blur it with broader constructs such as system justification, beliefs about mobility, or support for equal opportunity (Batruch et al., 2022; Darnon, Wiederkehr, et al., 2018; Deng & Wang, 2025; Jost et al., 2004; McCoy & Major, 2007). As a result, conceptually distinct orientations are often collapsed into a single indicator, and the distinction between descriptive perceptions of how rewards are allocated and normative preferences about how they should be allocated is acknowledged but rarely implemented analytically. A further limitation concerns the treatment of non-meritocratic factors such as family wealth, social connections, or inherited advantage. As Castillo et al. (2023) notes, many studies ask respondents to rate the importance of meritocratic and non-meritocratic factors for "getting ahead", but then collapse them into a single score. This is especially problematic when researchers treat both as opposite ends of the same continuum, for example, through subtraction scores (as in the Reynolds & Xian, 2014 approach). Such measures impose a zero-sum assumption—namely, that endorsing merit implies rejecting structural or relational advantages—and therefore rule out, by construction, the possibility that both beliefs coexist. Recent debates and empirical evidence suggest the opposite: support for merit can coexist with recognition of privilege, contacts, or structural constraints (Kwon & Pandian, 2024; C. Liu & Wang, 2025; Mijs, 2026; Tang et al., 2025; Zhu, 2025). The implication is clear: unidimensional indices, especially subtraction scores, can obscure substantively distinct belief profiles.

Responding to these limitations, Castillo et al. (2023) proposed a multidimensional framework for conceptualizing and measuring meritocratic beliefs in survey research. This proposal decomposes meritocratic beliefs along two analytically independent axes: perceptions versus preferences, and meritocratic versus non-meritocratic allocation principles. Preferences capture normative ideals about how rewards should be distributed, whereas perceptions capture descriptive evaluations of how society actually works (Janmaat, 2013). At the same time, meritocratic elements (e.g., effort, talent) and non-meritocratic elements (e.g., social origins, inherited privilege, networks) are treated as distinct components rather than opposite poles of a single continuum. This yields four non-redundant constructs—meritocratic preferences, meritocratic perceptions, non-meritocratic preferences, and non-meritocratic perceptions—and allows combinations that older measures collapse, such as endorsing meritocracy as an ideal while recognizing the role of family background and connections. The framework is also parsimonious and therefore well suited to large-scale surveys, although its reliance on only two items per factor may constrain reliability and content coverage.

Evidence from adult samples using confirmatory factor analysis suggests that respondents can differentiate among these four dimensions and that they are systematically related. In particular, stronger perceptions that non-meritocratic factors are rewarded tend to coexist with stronger preferences for meritocratic allocation (Castillo et al., 2023). This pattern aligns with Zhu's (2025) notion of "dual consciousness," understood here as the coexistence of meritocratic and non-meritocratic beliefs, as well as descriptive assessments ("is") and legitimacy judgments ("should").

2.2 The socialization of meritocracy at school

Socialization refers to the process through which individuals internalize norms, values, and practices, acquiring the dispositions needed to function in society. From a sociological perspective, this process is neither neutral nor purely individual, but structured by relations of power and inequality (Bourdieu, 2007). Likewise, social learning approaches emphasize that children adopt behavioral patterns and value systems modeled in their everyday environments, especially within families (Bandura, 1969). The internalization of meritocratic beliefs—such as the idea that effort and talent determine success—should therefore be understood not as a purely rational judgment but as the outcome of structured, socially differentiated socialization processes.

Families and schools are central socialization agencies of meritocratic beliefs. Parenting often reflects class-based strategies of status reproduction (Bernstein, 2005; Bourdieu & Passeron, 1990), and children internalize parental views of success, fairness, and responsibility through everyday interactions (Bandura, 1969). As far as schools are concerned, they are key sites of meritocratic socialization, making adolescence a crucial period for studying how these beliefs are formed. They promote narratives of effort, achievement, self-improvement, and education as legitimate routes to mobility (Chauvin et al., 2026; Darnon, Wiederkehr, et al., 2018; Mijs, 2016b), while embedding these ideals in routine practices of evaluation and classification such as grading, tracking, selection, awards, and credentials (Darnon et al., 2023; Resh & Sabbagh, 2017; Traini, 2022). At the same time, schools are also settings in which non-meritocratic or structural advantages—such as family resources, cultural capital, peer environments, and unequal school quality—become increasingly visible and consequential as students gain experience and knowledge (Goudeau & Cimpian, 2021; Goudeau & Croizet, 2017; Tang et al., 2025). A growing literature shows that considering meritocratic and non-meritocratic elements shed lights on internally differentiated orientations. Students may endorse meritocracy as a normative ideal while simultaneously recognizing structural constraints, a pattern described as "dual consciousness" (C. Liu & Wang, 2025; Tang et al., 2025).

Belief in school meritocracy has consequences both within schools and beyond them. In educational settings, meritocratic framing encourages students to interpret success and failure in individual terms, reinforcing personal responsibility while downplaying structural conditions (Goudeau & Cimpian, 2021; Lampert, 2013). Even disadvantaged students often adopt meritocratic narratives, suggesting that prolonged exposure to school-based evaluation shapes beliefs about justice and inequality (Henry & Saul, 2006; Tang et al., 2025). These beliefs also shape educational practices: among teachers, belief in school meritocracy is associated with greater support for competitive rather than cooperative practices, while among parents it predicts lower support for interventions aimed at reducing socioeconomic inequalities in school achievement (Darnon et al., 2023; Darnon, Smeding, et al., 2018). Beyond school, perceiving schools as meritocratic has been linked to greater legitimation of inequality, including stronger system justification, lower support for redistribution, and greater acceptance of unequal access to social services (Batruch et al., 2022; Castillo et al., 2024; Darnon, Smeding, et al., 2018; Wiederkehr et al., 2015). School meritocracy thus shapes not only educational trajectories, but also broader orientations that may normalize social inequality.

However, the measurement of meritocratic beliefs has received limited attention in school-based research. A key exception is the Belief in School Meritocracy (BSM) scale developed by Wiederkehr et al. (2015), an eight-item measure focused mainly on effort and equality of opportunity, which has been widely used in subsequent studies (Batruch et al., 2022; Darnon et al., 2023; Darnon, Smeding, et al., 2018; Darnon, Wiederkehr, et al., 2018). For example, Darnon, Smeding, et al. (2018) used the same items to measure both perceptions and preferences regarding school meritocracy in a French sample, showing that perceived school meritocracy—but not preferred school meritocracy—predicts lower support for more egalitarian and inclusive educational interventions. Although this represents an important step forward, the approach remains limited because it uses identical item wording for perceptions and preferences, excludes non-meritocratic principles, and does not test these distinctions within an explicit factorial measurement model. Another recent exception is Chauvin et al. (2026), who proposes a multidimensional measure of school meritocracy; however, it relies on a single scale that conflates beliefs about school and social meritocracy, focuses only on teachers, and remains restricted to perceived meritocracy. To date, no study has examined whether students differentiate between perceptions and preferences regarding both meritocratic and non-meritocratic principles, let alone whether these dimensions are stable across cohorts or over time, thereby limiting our understanding of their developmental trajectory and comparability.

2.3 This study

This study examines how students understand meritocracy during a formative stage of socialization—schooling—and whether these understandings can be measured reliably and comparably. Building on Castillo et al.'s (2023) conceptual and measurement framework, we test whether students differentiate between perceptions of how meritocracy operates ("what is") and preferences about how it should operate ("what ought to be"), and whether these judgments distinguish meritocratic elements (effort, talent) from non-meritocratic ones (family wealth, personal contacts). Beyond factorial validity, the paper focuses on measurement stability—whether the same latent constructs are captured comparably across school stages and over time within the same students.

Establishing measurement invariance is essential because it provides the empirical foundation for valid comparisons (Chen, 2007; Putnick & Bornstein, 2016). Without it, observed differences between groups or across time may reflect changes in how the instrument functions rather than substantive differences in the underlying constructs (Putnick & Bornstein, 2016; Van De Schoot et al., 2015). In sociological terms, invariance matters because comparisons are only interpretable if a given latent score carries the same meaning across ages or time points; otherwise, apparent developmental or group differences may conflate substantive variation with measurement artifacts (Y. Liu et al., 2017; Van De Schoot et al., 2012).

The study uses longitudinal data from two cohorts of Chilean secondary students: 8th graders (approximately 12–13 years old) and 10th graders (approximately 15–16 years old) at baseline, followed across two waves. These cohorts are theoretically strategic because they capture distinct stages of adolescence, when reasoning about justice and inequality becomes more sophisticated and school-based sorting mechanisms—grades, tracking, and peer comparisons—become increasingly salient (Henry & Saul, 2006; Resh & Sabbagh, 2014). Developmental research further suggests that by around age 10, children can already articulate judgments about social differences, often emphasizing individual, merit-based explanations while beginning to incorporate opportunity-based accounts (Imhoff & Brussino, 2025; Sigelman, 2013). As students move through the school system, cumulative exposure to evaluation and selection may make the limits of effort as a sufficient explanation for success more visible, generating a potential "reality shock" as meritocratic discourse clashes with experienced inequality and constraint (Tang et al., 2025). Both cohorts are embedded in the same highly stratified educational system, yet differ in their cumulative exposure to schooling and to the meritocratic narratives the system promotes. We therefore test invariance across cohorts and waves to determine whether observed differences and changes can be interpreted substantively rather than as artifacts of measurement non-equivalence.

2.3.1 Hypotheses

H_1 **(Dimensional differentiation)**. Students' meritocratic beliefs are best represented by four distinct factors: perceived meritocracy, perceived non-meritocracy, preferred meritocracy, and preferred non-meritocracy.

H_2 **(Between-cohort measurement invariance)**. The four-factor model of meritocratic beliefs is invariant across age cohorts.

H_3 **(Longitudinal measurement invariance)**. The four-factor model of meritocratic beliefs is invariant across waves within students.

3 Method

3.1 Participants

This study uses secondary data from the 2023 and 2024 waves of the Panel Survey on Education and Meritocracy (EDUMER), which examines school-age students' beliefs, attitudes, and behaviors regarding meritocracy, inequality, and citizenship. The questionnaire was administered to sixth-grade and first-year secondary students in nine schools in Chile's Metropolitan and Valparaíso regions. Although the sample was non-probabilistic and lacked quota controls, a minimum target of 900 students was set to ensure adequate statistical power. After data processing and listwise deletion, the analytical sample included 846 students in Wave 1 (386 girls, 421 boys, 39 identifying as other; M_{age} = 13.4, SD_{age} = 1.6) and 662 in Wave 2 (303 girls, 338 boys, 21 identifying as other; M_{age} = 14.4, SD_{age} = 1.6), yielding a retention rate of 78.3% (attrition = 21.7%). The main sources of attrition were school transfers between academic years and high absenteeism in several schools.

Table 1: Items for perceptions and preferences for meritocracy scale

Components	Dimensions	Item (English)	Item original (Spanish)
Perception	Meritocratic	In Chile people are rewarded for their efforts	En Chile las personas son recompensadas por sus esfuerzos
		In Chile people are rewarded for their intelligence and ability	En Chile las personas son recompensadas por su inteligencia y habilidad
	Non meritocratic	In Chile those with wealthy parents do much better in life	En Chile a quienes tienen padres ricos les va mucho mejor en la vida
		In Chile those with good contacts do much better in life	En Chile a quienes tienen buenos contactos les va mejor en la vida
Preference	Meritocratic	Those who work harder should reap greater rewards than those who work less hard	Quienes más se esfuerzan deberían obtener mayores recompensas que quienes se esfuerzan menos
		Those with more talent should reap greater rewards than those with less talent	Quienes poseen más talento deberían obtener mayores recompensas que quienes poseen menos talento
	Non meritocratic	It is good that those who have rich parents do better in life	Está bien que quienes tengan padres ricos les vaya mejor en la vida
		It is good that those who have good contacts do better in life	Está bien que quienes tengan buenos contactos les vaya mejor en la vida

3.2 Procedure

Data were collected by a professional research firm using a Computer-Assisted Web Interviewing (CAWI) approach, based on online questionnaires. All participants received informed consent, which was reviewed and approved by a parent or legal guardian. To enable longitudinal linkage while protecting confidentiality, the survey firm generated anonymous unique identifiers for each student, and no directly identifying information was included in the analytic files.

3.3 Measures

3.3.1 Scale of perceptions and preferences of meritocracy

Meritocratic and non-meritocratic perceptions and preferences were measured using the same items proposed in the original scale (Castillo et al., 2023). Each of the four dimensions of the scale is measured by two items. The dimensions are:

- Perceived meritocracy: the extent to which effort and ability are rewarded in Chile,
- Perceived non-meritocracy: the extent to which success is perceived as linked to connections and family wealth,
- Preference for meritocracy: agreement with that those who work harder or are more talented should be better rewarded,
- Preference for non-meritocracy: agreement with that it is acceptable for individuals with better connections or wealthy parents to achieve greater success (see Table 1).

Each item is rated on a four-point Likert scale ranging from “strongly disagree” (1) to “strongly agree” (4), therefore higher scores indicate stronger endorsement of the corresponding perception or preference.

Table 2: Descriptive statistics for cohort level

Wave	Cohort level	N	Age		Gender		
			Mean	SD	Man	Women	Others
Wave 1	Primary	406	11.85	0.56	195	191	20
	Secondary	440	14.79	0.80	226	195	19
	Total	846	13.38	1.63	421	386	39
Wave 2	Primary	319	12.83	0.55	161	145	13
	Secondary	343	15.76	0.75	177	158	8
	Total	662	14.35	1.61	338	303	21

3.3.2 Cohort level

To distinguish students' academic level across waves for conditional and multi-group invariance analyses, we created a cohort indicator capturing whether the respondent belonged to the primary (sixth grade) or secondary (first-year secondary) cohort at the time of each survey wave. Descriptive statistics by cohort and wave are reported in Table 2.

3.4 Analytical strategy

3.4.1 Measurement model and estimation

To evaluate the underlying structure of the scale, we employed Confirmatory Factor Analysis (CFA) based on a measurement model with four latent factors, using Diagonally Weighted Least Squares with robust correction (WLSMV) estimation. This estimator is particularly suitable for ordinal data, such as four-point Likert-type scales, as it avoids the bias associated with treating categorical data as continuous (Kline, 2023).

Model fit was assessed following the guidelines of Brown (2015), using several indices: the Comparative Fit Index (CFI) and the Tucker-Lewis Index (TLI), with acceptable values above 0.95; the Root Mean Square Error of Approximation (RMSEA), with values below 0.06 indicating good fit; and the Chi-square statistic (acceptable fit indicated by $p > 0.05$ and a Chi-square/df ratio < 3).

3.4.2 Measurement invariance

A key contribution of this study is its assessment of measurement stability through factorial invariance testing (Putnick & Bornstein, 2016). We implemented two complementary strategies: longitudinal invariance across two panel waves and cross-group invariance across school levels (primary vs. secondary students). Because the items are ordinal, we followed recent recommendations for ordered categorical indicators, specifically the longitudinal approach of Y. Liu et al. (2017) and the cross-group procedure of Svetina et al. (2020). In the longitudinal analyses, we estimated the conventional hierarchy of configural, metric, scalar, and strict models, progressively constraining factor loadings, thresholds/intercepts, and residual variances over time. In the cross-group analyses, however, we followed the more appropriate three-step sequence for ordinal indicators proposed by Svetina et al. (2020), estimating a configural model, then a threshold-invariant model, and finally a model jointly constraining thresholds and factor loadings across groups.

In addition to the chi-square difference test, we assessed invariance using changes in approximate fit indices. Following Chen (2007), we treated ΔCFI as the primary criterion and ΔRMSEA as supplementary, indicating noninvariance when $\Delta \text{CFI} \geq -0.010$ and $\Delta \text{RMSEA} \geq 0.015$.

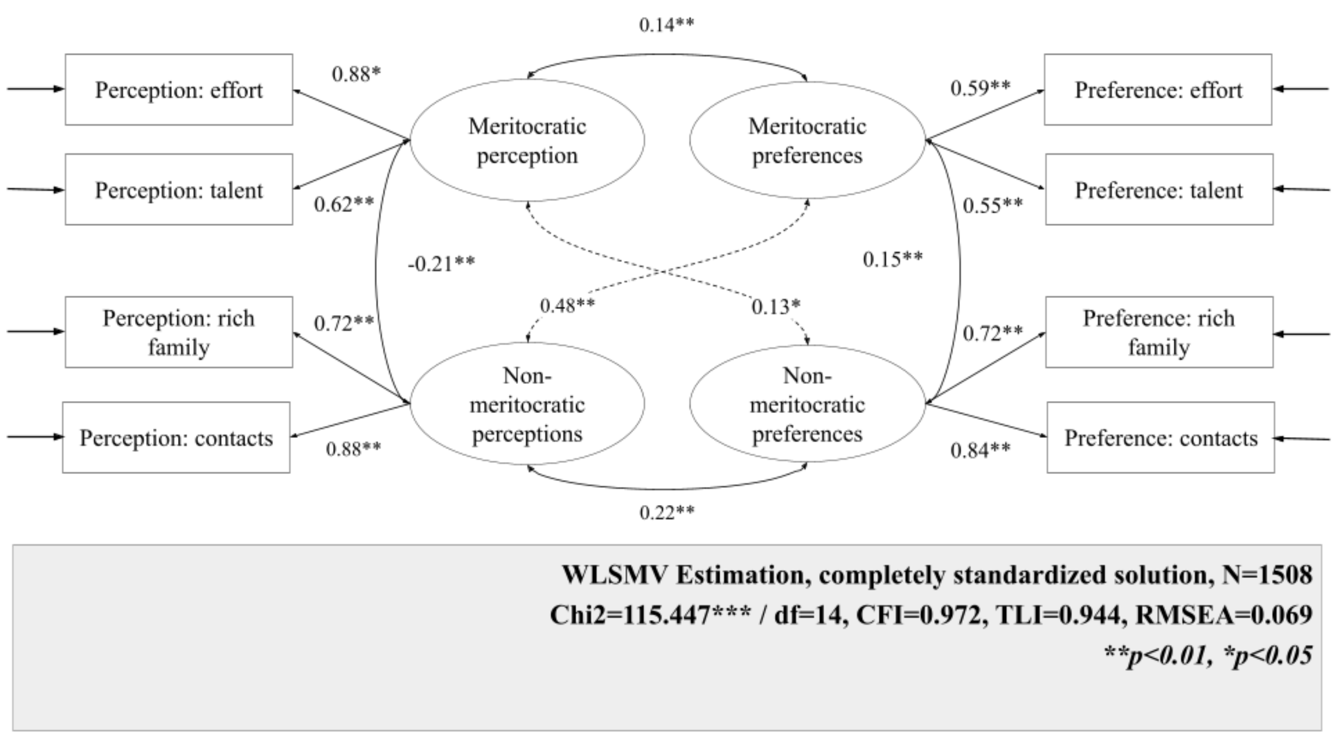
All analyses were performed using the *lavaan* package in R version 4.5.2. The hypotheses of this research were pre-registered on the Open Science Framework (OSF).

4 Results

4.1 Measurement model

Figure 1 presents the standardized factor loadings and the overall fit indices of the general measurement model estimated on the pooled analytical sample across cohorts and waves. The model shows a reasonable, though not fully satisfactory, fit to the data (CFI = 0.97; TLI = 0.94; RMSEA = 0.07). Although the CFI falls within commonly accepted ranges, the RMSEA and TLI remain below conventional thresholds, indicating that fit is acceptable in some respects but not uniformly strong across indices. Overall, these results suggest that the proposed four-factor structure captures the main covariance pattern in the combined sample reasonably well [1].

Figure 1: Multigroup model diagram



All standardized factor loadings are statistically significant and generally moderate to strong, though their magnitude varies across dimensions. For meritocratic perceptions, both indicators load positively on the latent factor, with effort showing a particularly strong loading (β = 0.88) and talent a moderately strong loading (β = 0.62). Meritocratic preferences exhibit

[1]Factor loadings and fit indices for the separate analytical samples (i.e., per wave and cohort level) are available in the Supplementary Material.

more even, but somewhat weaker, loadings for effort (β = 0.59) and talent (β = 0.55). For non-meritocratic perceptions, both indicators load strongly, especially contacts (β = 0.88), followed by coming from a rich family (β = 0.72). Non-meritocratic preferences also show strong and relatively balanced loadings, with contacts (β = 0.84) and rich family (β = 0.72), indicating that both indicators are similarly representative of this latent dimension.

The latent correlations reveal a differentiated structure among the four dimensions. Meritocratic perceptions and meritocratic preferences are positively correlated (r = 0.14), consistent with some alignment between what students perceive is rewarded and what they believe should be rewarded. Meritocratic and non-meritocratic perceptions are negatively correlated ($r = -0.21$), suggesting that students who perceive greater meritocracy in society tend to view a weaker role for connections and family wealth. At the same time, non-meritocratic perceptions are positively related to meritocratic preferences (r = 0.48, strongest correlation), suggesting that perceiving society as rewarding contacts and wealth may coincide with stronger endorsement of meritocratic ideals. Finally, meritocratic and non-meritocratic preferences are modestly positively correlated (r = 0.15), and non-meritocratic perceptions and preferences are also positively associated (r = 0.22), indicating that recognizing non-meritocratic mechanisms in practice is linked to a greater acceptance of their legitimacy, albeit to a moderate extent.

4.2 Invariance analysis

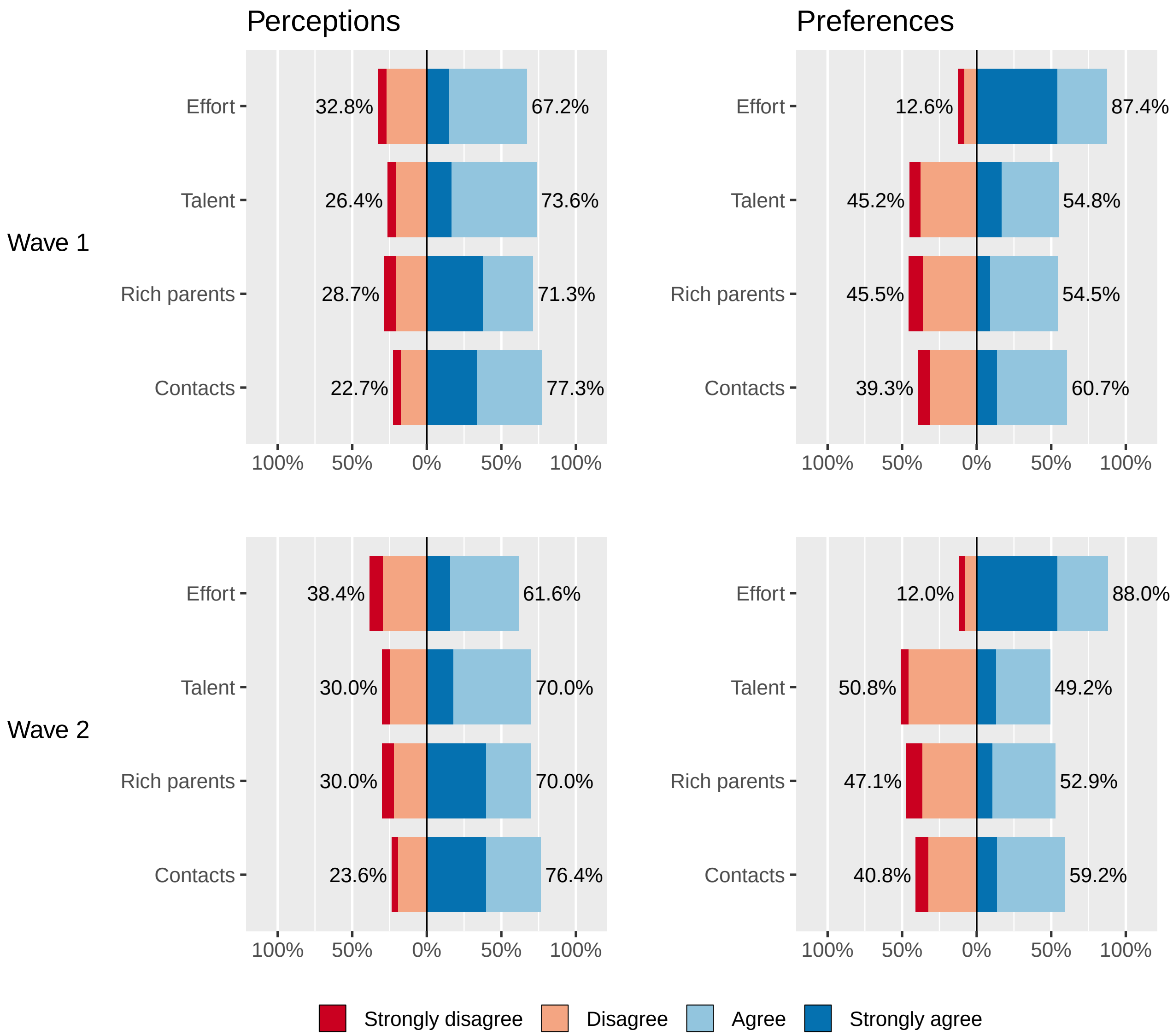
4.2.1 Longitudinal invariance

Figure 2 displays the response distributions for the meritocracy scale, distinguishing perceptions (Panel A) from preferences (Panel B) across waves. Across both waves, a clear and stable gap emerges between how students perceive social rewards to operate and how they think they should operate. In the perception items, agreement is consistently high for contacts, parental wealth, and talent, whereas effort is also widely endorsed but to a lesser extent, indicating that students recognize both meritocratic and non-meritocratic determinants of success and view talent as more influential than effort. In the preference items, by contrast, effort is overwhelmingly endorsed as the legitimate basis for rewards, far exceeding its perceived role in practice. Talent occupies a more ambivalent normative position, parental wealth remains evenly divided, and contacts are more often accepted than rejected, suggesting that non-meritocratic criteria are not uniformly seen as illegitimate. Wave 2 closely replicates these patterns, with only modest shifts. Overall, the descriptive results reveal a persistent paradox: strong endorsement of effort-centered meritocratic ideals alongside skepticism about their realization in practice.

Table 3 reports the results of the longitudinal invariance tests. Overall, the scale achieved the strongest level of invariance (strict invariance) across the two waves. The strict model showed good fit, $\chi^2(84)$ = 130.17, CFI = 0.991, RMSEA = 0.031 (90% CI: 0.020–0.040). Moreover, compared with the strong (scalar) model, the deterioration in fit was negligible: $\Delta\chi^2(4)$ = 1.635, ΔCFI = 0.000, and ΔRMSEA = -0.002. These differences fall well within commonly used cutoffs for invariance evaluation (Chen, 2007), indicating that constraining residual variances in addition to loadings and thresholds/intercepts is tenable. Substantively, this implies that the item–factor relations are stable over time (metric invariance) and that, for a given level of the latent trait, respondents are expected to endorse the same response categories across waves (scalar invariance), supporting the comparability of scores and latent parameters within students over time.

As an additional diagnostic, we used univariate score tests to identify potential sources of localized misfit at each step of the invariance hierarchy. The tests revealed no meaningful violations of the imposed equality constraints, suggesting that model

Figure 2: Distribution of responses on the scale of perceptions and preferences regarding meritocracy by wave



Source: own elaboration based on EDUMER Panel Survey (N Wave 1 = 846; N Wave 2 = 662)

Table 3: Longitudinal invariance results

Model	χ^2 (df)	CFI	RMSEA (90% CI)	$\Delta\chi^2$ (Δdf)	ΔCFI	ΔRMSEA	Decision
Configural	117.7 (68)	0.991	0.035 (0.024-0.046)	0 (0)	0	0.000	Reference
Weak	122.51 (72)	0.990	0.034 (0.024-0.045)	4.809 (4)	0	-0.001	Accept
Strong	128.53 (80)	0.991	0.032 (0.021-0.042)	6.02 (8)	0	-0.002	Accept
Strict	130.17 (84)	0.991	0.031 (0.02-0.04)	1.635 (4)	0	-0.002	Accept

Table 4: Cohort (multigroup) invariance results

Model	χ^2 (df)	CFI	RMSEA (90% CI)	$\Delta\chi^2$ (Δdf)	ΔCFI	ΔRMSEA	Decision
Configural	24.95 (26)	0.993	0.036 (0.009-0.057)	0 (0)	0.000	0.000	Reference
Tresholds	57.56 (42)	0.980	0.049 (0.033-0.064)	43.875 (16) ***	-0.013	0.013	Reject
Tresholds + Loadings	72.35 (46)	0.973	0.053 (0.039-0.067)	17.075 (4) **	-0.006	0.005	Reject

fit was not driven by a small subset of parameters. Global score tests and the smallest p-values at each step are reported in the Supplementary Material.

4.2.2 Cohort invariance

Figure 3 shows response distributions by cohort, separating perceptions and preferences. In primary education, effort and talent are perceived as similarly rewarded, while contacts and family wealth are also seen as advantageous, though with more disagreement. In secondary education, perceptions become less meritocratic: agreement that effort is rewarded declines, whereas contacts and family wealth are seen as more strongly shaping success. Preferences are more stable across cohorts. In both groups, effort is the most strongly endorsed legitimate basis for reward, while talent, family wealth, and especially contacts remain more ambivalent. Notably, contacts are more often accepted than rejected in both cohorts. Overall, the cohort comparison reveals a widening gap between an effort-centered meritocratic ideal and the perceived importance of non-meritocratic advantages in practice.

Table 4 summarizes the cross-cohort invariance tests. The scale achieves configural invariance between primary and secondary students, but not threshold invariance, and therefore does not meet the prerequisites for stronger forms of invariance (Svetina et al., 2020; Wu & Estabrook, 2016). Constraining thresholds worsened model fit (ΔCFI = −.013; ΔRMSEA = .013), exceeding recommended cutoffs (Chen, 2007). This indicates that response categories do not map equivalently onto the latent constructs across cohorts, limiting direct comparisons of latent means or other latent differences.

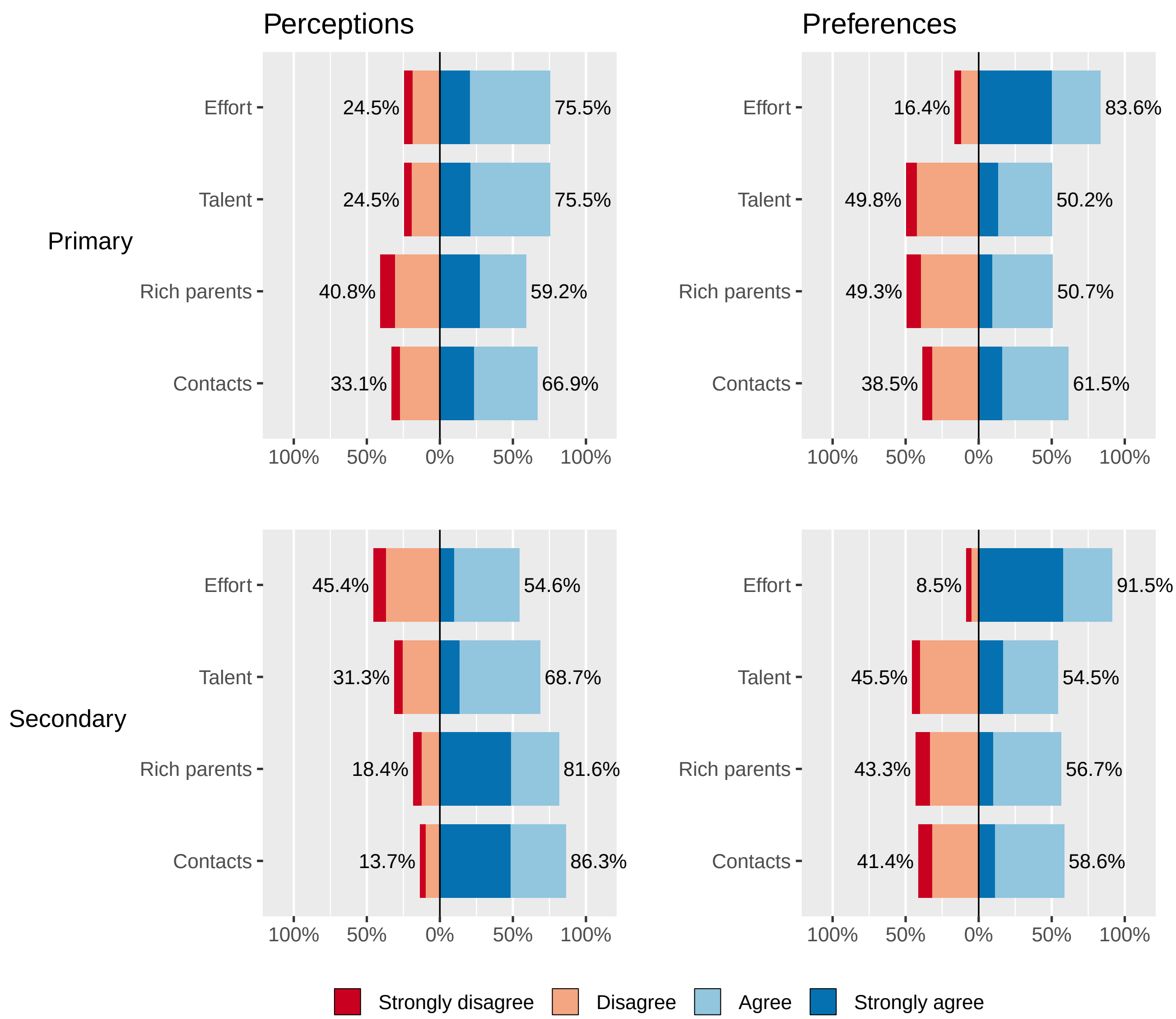
To locate the source of cross-cohort non-invariance, we inspected localized misfit in the threshold-invariant model. The largest violations concerned the perceived-effort item, especially the second threshold ($\chi^2 = 10.12, p = .001$) and, to a lesser extent, the first ($\chi^2 = 7.55, p = .006$), suggesting that older students may apply a more critical standard when judging whether effort is rewarded in society.

Despite the lack of cohort invariance, re-estimating the longitudinal invariance models with cohort as a covariate predicting each latent factor did not materially change model fit or conclusions, suggesting that invariance over time remains tenable. These robustness checks are reported in the Supplementary Material.

5 Discussion

The first hypothesis proposed that students' meritocratic beliefs would be best represented by four distinct factors: perceived meritocracy, perceived non-meritocracy, preferred meritocracy, and preferred non-meritocracy. The confirmatory factor anal-

Figure 3: Distribution of responses on the scale of perceptions and preferences regarding meritocracy by cohort



Source: own elaboration based on EDUMER Panel Survey (N Primary = 725; N Secondary = 783) yses support this expectation. Consistent with evidence from adult populations (Castillo et al., 2023), students already distinguish between perceptions and preferences, and between meritocratic and non-meritocratic principles at an early stage of school socialization.

These findings contribute to debates on how meritocratic beliefs should be conceptualized and measured (Kwon & Pandian, 2024; Mijs, 2026; Zhu, 2025). Perceptions of "what is" and preferences about "what should be" are empirically separable among school-aged students, and meritocratic and non-meritocratic principles coexist rather than forming opposite poles of a single continuum. This directly challenges zero-sum measurement strategies, such as subtraction scores, that assume endorsing merit implies rejecting privilege (Reynolds & Xian, 2014; Wiesner & Sachweh, 2026). Instead, our results are more consistent with recent accounts arguing that recognition of non-meritocratic constraints may increase without displacing merit ideals (C. Liu & Wang, 2025; Tang et al., 2025). The positive association between perceived non-meritocracy and stronger meritocratic preferences supports this interpretation, suggesting a normative-reactive response to perceived unfairness (cf. Castillo et al., 2023).

The findings also align with research on meritocracy in educational settings. Schools circulate meritocratic narratives of effort, achievement, and mobility while institutionalizing merit through grading, selection, and tracking (Darnon et al., 2023; Mijs, 2016b; Resh & Sabbagh, 2017; Traini et al., 2025). Our results extend this literature by showing that students differentiate not only between perceptions and preferences but also between meritocratic and non-meritocratic principles. Rather than focusing on school meritocracy itself, the study shifts attention to how students understand social meritocracy within school contexts, while offering a basis for developing measures better suited to these settings. More specifically, the findings reinforce earlier evidence that students can distinguish meritocratic elements in both perceptions and preferences (Darnon, Smeding, et al., 2018), and add that they can also differentiate meritocratic from non-meritocratic principles in both domains. Because school meritocracy has been linked to inequality legitimation and broader distributive attitudes (Batruch et al., 2022; Castillo et al., 2024; Darnon, Wiederkehr, et al., 2018; Wiederkehr et al., 2015), an important implication is that meritocracy effects are likely dimension-specific: perceptions are not equivalent to preferences, and meritocratic preferences need not have the same implications as non-meritocratic ones.

The second hypothesis stated that the measurement model would be stable over time, and the results support this expectation. Achieving strict longitudinal invariance indicates that the scale functions equivalently across waves, allowing within-student differences over time to be interpreted as substantive rather than as artifacts of item interpretation or response use (Y. Liu et al., 2017; Putnick & Bornstein, 2016; Van De Schoot et al., 2015). This strengthens longitudinal inference and provides a firmer basis for studying developmental patterns in meritocratic beliefs, something rarely established in prior research (e.g., Chauvin et al., 2026; C. Liu & Wang, 2025; Tang et al., 2025; Wiederkehr et al., 2015).

The third hypothesis proposed invariance across educational levels, but the results did not support this expectation. The lack of cross-level invariance appears to be concentrated in perceived meritocracy, suggesting that students at different school stages interpret whether effort is rewarded differently (Putnick & Bornstein, 2016; Van De Schoot et al., 2012). Substantively, primary students perceive greater rewards for effort than secondary students do. A plausible explanation is that cumulative exposure to grading, selection, tracking, and broader social inequalities produces a "reality shock" in which effort appears less sufficient for success (Tang et al., 2025). This means that differences across school stages may reflect not only different levels of meritocratic belief, but also different meanings attached to meritocratic reward.

6 Conclusion

Meritocratic beliefs remain central to how inequality is interpreted, and schools are key settings in which they are produced and reinforced through evaluation, tracking, and selection (Darnon et al., 2023; Mijs, 2016b; Resh & Sabbagh, 2017; Traini et al., 2025). However, research on adolescents' meritocratic beliefs has often relied on unidimensional measures that conflate perceptions with preferences and treat meritocratic and non-meritocratic principles as opposites. Extending Castillo et al.'s (2023) framework to adolescents, this study tests whether students distinguish between perceived and preferred meritocracy and non-meritocracy, whether merit endorsement can coexist with recognition of non-meritocratic advantage, and whether this model is stable across school stages and over time in Chile.

Our results show that meritocratic beliefs can be represented as distinct dimensions already at school age, much like in adult populations. Adolescents differentiate between perceptions and preferences, and between meritocratic and non-meritocratic principles. This multidimensional approach also shows that endorsement of merit can coexist with recognition of non-meritocratic advantage. These findings support measuring meritocracy as a set of related but distinct dimensions rather than collapsing them into a single score, especially through zero-sum indices. This is important because what is often interpreted in the literature as a change in belief in meritocracy may instead reflect increasing recognition of non-meritocratic constraints rather than a straightforward decline in merit endorsement, consistent with recent education-focused accounts of dual consciousness (C. Liu & Wang, 2025; Tang et al., 2025; Zhu, 2025).

The multidimensional scale of perceived and preferred meritocracy and non-meritocracy has important methodological implications for the study of meritocratic beliefs at schools. Its strict longitudinal invariance indicates that the instrument functions equivalently over time for individual students, providing a robust basis for longitudinal analysis and allowing within-student change to be interpreted as substantive rather than a measurement artifact. However, the scale does not achieve invariance across cohorts, so comparisons across educational stages should be made with caution. A plausible interpretation is that students at different stages of schooling differ in how they judge whether effort is rewarded in society, possibly because cumulative exposure to schooling makes effort appear less sufficient for success among older students (Tang et al., 2025). This interpretation, however, remains tentative and calls for further research.

Despite its contributions, this study has limitations that point to future research. The scale is deliberately minimalist (as designed for survey studies), with only two items per factor, which limits content coverage and reliability. The design also includes only two waves and two school levels, restricting the analysis of longer developmental trajectories. In addition, cross-level non-invariance suggests the need to refine perceived-meritocracy items and test for differential item functioning. Beyond measurement, future research should examine the substantive implications of these beliefs in school settings, including their links to students' sense of injustice, academic motivation and performance, attitudes toward selection and tracking, and broader views about inequality and educational opportunity. It should also examine how other members of school communities, such as teachers and parents, understand meritocracy and how their views interact with students' own beliefs and behaviors (e.g., Darnon et al., 2023). Such work would help clarify how meritocratic beliefs are reproduced and negotiated within school communities, and how they shape students' educational experiences and legitimize inequality in school contexts.

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